Applied Statistics (MATH10096)

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Semester 2, 2024/2025

Linear model theory (Section 4 of the lecture notes)

QR decomposition

- The function `lm` uses the QR decomposition behind the scenes. Let us then learn this method!

- As with any real matrix, the design matrix $\mathbf{X}$ can always be decomposed as

$$\mathbf{X} = \mathcal{Q}\begin{pmatrix} \mathbf{R} \\ \mathbf{0} \end{pmatrix} = \mathbf{QR}$$

- Here we have that:

  - $\mathbf{R}$ is a $p \times p$ upper triangular matrix.

  - $\mathbf{0}$ is an $(n-p) \times p$ matrix of zeroes.

  - $\mathcal{Q}$ is an $n \times n$ orthogonal matrix, the first p columns of which form $\mathbf{Q}$ (whose dimension is $n \times p$).

Linear model theory (Section 4 of the lecture notes)

QR decomposition

$\hookrightarrow$ Multiplying the vector $\mathbf{y} - \mathbf{X}\beta$ by $\mathcal{Q}^{\mathsf{T}}$ does not change its length and it implies that

$$\|\mathbf{y} - \mathbf{X}\beta\|^2 = \|\mathcal{Q}^{\mathsf{T}}\mathbf{y} - \mathcal{Q}^{\mathsf{T}}\mathbf{X}\beta\|^2 = \left\|\mathcal{Q}^{\mathsf{T}}\mathbf{y} - \mathcal{Q}^{\mathsf{T}}\mathcal{Q}\begin{pmatrix}\mathbf{R}\\ \mathbf{0}\end{pmatrix}\beta\right\|^2 = \left\|\mathcal{Q}^{\mathsf{T}}\mathbf{y} - \begin{pmatrix}\mathbf{R}\\ \mathbf{0}\end{pmatrix}\beta\right\|^2$$

$\hookrightarrow$ Defining $\mathbf{f}$ as a $p \times 1$ dimensional vector and $\mathbf{r}$ as a $(n-p) \times 1$ vector, so that $\begin{pmatrix}\mathbf{f}\\ \mathbf{r}\end{pmatrix} \equiv \mathcal{Q}^{\mathsf{T}}\mathbf{y}$, yields

$$\|\mathbf{y} - \mathbf{X}\beta\|^2 = \left\|\begin{pmatrix}\mathbf{f}\\ \mathbf{r}\end{pmatrix} - \begin{pmatrix}\mathbf{R}\\ \mathbf{0}\end{pmatrix}\beta\right\|^2 = \|\mathbf{f} - \mathbf{R}\beta\|^2 + \|\mathbf{r}\|^2$$

$\hookrightarrow$ Recall that our goal is to find the vector β that minimises $\|\mathbf{y} - \mathbf{X}\beta\|^2$ or, equivalently, the vector β that minimises $\|\mathbf{f} - \mathbf{R}\beta\|^2 + \|\mathbf{r}\|^2$.

$\hookrightarrow$ The length of $\mathbf{r}$ does not depend on β and $\|\mathbf{f} - \mathbf{R}\beta\|^2$ can be reduced to zero by choosing β so that $\mathbf{R}\beta$ equals $\mathbf{f}$.

↪ That is, the least squares estimator satisfies

$$\mathbf{R}\hat{\beta} = \mathbf{f}. \tag{1}$$

↪ Provided that $\mathbf{X}$ and therefore $\mathbf{R}$ have full column rank, we have

$$\hat{\beta} = \mathbf{R}^{-1}\mathbf{f} = \mathbf{R}^{-1}[\mathcal{Q}^\top \mathbf{y}]_{1:p} = \mathbf{R}^{-1}\mathbf{Q}^\mathsf{T}\mathbf{y}.$$

↪ Note that because $\mathbf{R}$ is an upper triangular matrix, the system of equations in (1) can be easily solved by back substitution, thus allowing to easily find $\hat{\beta}$ without the need to invert any matrix.

↪ Further note that inverting $\mathbf{R}$ is numerically more stable than inverting $\mathbf{X}^\mathsf{T}\mathbf{X}$.

↪ Notice that

$$||\mathbf{y} - \mathbf{X}\hat{\beta}||^2 = ||\mathbf{f} - \mathbf{R}\hat{\beta}||^2 + ||\mathbf{r}||^2 = ||\mathbf{f} - \mathbf{R}\mathbf{R}^{-1}\mathbf{f}||^2 + ||\mathbf{r}||^2,$$

and thus $||\mathbf{r}||^2 = ||\mathbf{y} - \mathbf{X}\hat{\beta}||^2$ is the residual sum of squares for the model fit.

Linear model theory (Section 4 of the lecture notes)

QR decomposition

- The two estimators of $\boldsymbol{\beta}$, based on the design matrix directly and based on the QR decomposition, are obviously equivalent.
- Of course, the same equivalence also holds for the covariance matrix. Note that the covariance matrix of $\mathbf{f} = \mathbf{Q}^{\mathsf{T}}\mathbf{y}$ is

$$\text{var}(\mathbf{f}) = \text{var}(\mathbf{Q}^{\mathsf{T}}\mathbf{y}) = \mathbf{Q}^{\mathsf{T}}\text{var}(\mathbf{y})\mathbf{Q} = \sigma^{\mathbf{2}}\mathbf{Q}^{\mathsf{T}}\mathbf{I_n}\mathbf{Q} = \sigma^{\mathbf{2}}\mathbf{I_p},$$

which implies that

$$\mathbf{V}_{\hat{\beta}} = \text{var}(\hat{\boldsymbol{\beta}}) = \text{var}(\mathbf{R}^{-1}\mathbf{f}) = \mathbf{R}^{-1}\text{var}(\mathbf{f})\mathbf{R}^{-\mathsf{T}} = \mathbf{R}^{-\mathbf{1}}\mathbf{R}^{-\mathsf{T}}\sigma^{\mathbf{2}}.$$

- Remember that we have found before that $\mathbf{V}_{\hat{\beta}} = (\mathbf{X}^{\mathsf{T}}\mathbf{X})^{-1}\sigma^2$. Finally, we only need to notice that

$$\begin{aligned}\mathbf{V}_{\hat{\beta}} &= (\mathbf{X}^{\mathsf{T}}\mathbf{X})^{-1}\sigma^2 = \left(\mathbf{R}^{\mathsf{T}}\mathbf{Q}^{\mathsf{T}}\mathbf{Q}\mathbf{R}\right)^{-1}\sigma^2 = \left(\mathbf{R}^{\mathsf{T}}\mathbf{R}\right)^{-1}\sigma^2 \\ &= \mathbf{R}^{-1}\mathbf{R}^{-\mathsf{T}}\sigma^2.\end{aligned}$$

↪ Let us revisit the simulated data example.

```
qrx <- qr(X)
R <- qr.R(qrx)
Q <- qr.Q(qrx)
betahat <- backsolve(R, crossprod(Q,y))
betahat
```

```
##            [,1]
## [1,] 0.9038372
## [2,] 1.0066440
## [3,] 0.9999622
## [4,] 1.0000001
```

```
lm_fit$coefficients
```

```
## XIntercept          Xx         Xx2         Xx3
##  0.9038373   1.0066440   0.9999622   1.0000001
```

- Confidence intervals and hypothesis tests are probabilistic concepts, so at this stage we need to turn our linear model into a fully probabilistic model, by specifying a full distribution for the ϵ_i, for $i = 1, \ldots, n$.

- We will simply assume that independently $\epsilon_i \sim N(0, \sigma^2)$ or, equivalently, that $\boldsymbol{\epsilon} \sim N(\mathbf{0}, \sigma^2\mathbf{I}_n)$ which, in turn, implies that $\mathbf{y} \sim N(\mathbf{X}\boldsymbol{\beta}, \mathbf{I}\sigma^2)$.

- Because $\hat{\boldsymbol{\beta}}$ is a linear transformation of $\mathbf{y}$, it follows that $\hat{\boldsymbol{\beta}}$ is also normally distributed

$$\hat{\boldsymbol{\beta}} \sim N(\boldsymbol{\beta}, \mathbf{V}_{\hat{\boldsymbol{\beta}}}), \text{ where } \mathbf{V}_{\hat{\boldsymbol{\beta}}} = \mathbf{R}^{-1}\mathbf{R}^{-\mathsf{T}}\sigma^2.$$

- This is not yet in the most directly usable form: it gives the distribution of $\hat{\boldsymbol{\beta}}$ in terms of the unknowns $\boldsymbol{\beta}$ and σ^2.

- As we have seen for the simple linear model, if we want to make inferences about β_j, for $j = 1, \ldots, p$, it is helpful if we can start from a pivotal statistic, i.e., a function of the data and unobservable parameters whose distribution does not depend on any unknown parameters.

- Let us first consider the distribution of $\mathcal{Q}^\mathsf{T}\mathbf{y}$. It is a linear transformation of the normal random vector $\mathbf{y}$ so it must also be a normal random vector.

- The covariance matrix of $\mathcal{Q}^\mathsf{T}\mathbf{y}$ is

$$\mathbf{V}_{\mathcal{Q}^\mathsf{T}\mathbf{y}} = \text{var}(\mathcal{Q}^\mathsf{T}\mathbf{y}) = \mathcal{Q}^\mathsf{T}\mathbf{I}\mathcal{Q}\sigma^2 = \mathbf{I}\sigma^2.$$

- Since, for the normal distribution only, zero covariance implies independence, the elements of $\mathcal{Q}^\mathsf{T}\mathbf{y}$ are mutually independent.

Linear model theory (Section 4 of the lecture notes)

Further inference results: $(\hat{\beta}_j - \beta_j)/\hat{\sigma}_{\hat{\beta}_j} \sim t_{n-p}$

↪ Furthermore

$$\mathbf{E}\begin{pmatrix}\mathbf{f}\\ \mathbf{r}\end{pmatrix} = \mathbf{E}(\mathcal{Q}^T\mathbf{y}) = \mathcal{Q}^T\mathbf{E}(\mathbf{y}) = \mathcal{Q}^T\mathbf{X}\boldsymbol{\beta} = \mathcal{Q}^T\mathcal{Q}\begin{pmatrix}\mathbf{R}\\ \mathbf{0}\end{pmatrix}\boldsymbol{\beta} = \begin{pmatrix}\mathbf{R}\\ \mathbf{0}\end{pmatrix}\boldsymbol{\beta}.$$

↪ This implies that

$$\mathbb{E}(\mathbf{f}) = \mathbf{R}\boldsymbol{\beta} \quad \text{and} \quad \mathbb{E}(\mathbf{r}) = \mathbf{0}_{n-p}.$$

↪ So we have that

$$\mathbf{f} \sim N(\mathbf{R}\boldsymbol{\beta}, \mathbf{I}_p\sigma^2) \;\; \text{and} \;\; \mathbf{r} \sim N(\mathbf{0}_{n-p}, \mathbf{I}_{n-p}\sigma^2),$$

with both vectors independent of each other.

Linear model theory (Section 4 of the lecture notes)

Further inference results: $(\hat{\beta}_j - \beta_j)/\hat{\sigma}_{\hat{\beta}_j} \sim t_{n-p}$

↪ Because the $n-p$ elements of $\mathbf{r}$ are i.i.d. $N(0, \sigma^2)$ random variables, we have that

$$\frac{1}{\sigma^2}\|\mathbf{r}\|^2 = \frac{1}{\sigma^2}\sum_{i=1}^{n-p} r_i^2 = \sum_{i=1}^{n-p}\left(\frac{r_i}{\sigma}\right)^2 \sim \chi^2_{n-p}.$$

↪ The mean of a χ^2_{n-p} random variable is $n-p$, so this result is sufficient (but not necessary; see Workshop sheet 1) to imply that

$$\hat{\sigma}^2 = \|\mathbf{r}\|^2/(n-p).$$

is an unbiased estimator of σ^2.

↪ The independence of the elements of $\mathbf{r}$ and $\mathbf{f}$ also implies that $\hat{\boldsymbol{\beta}}$ and $\hat{\sigma}^2$ are independent.

Linear model theory (Section 4 of the lecture notes)

Further inference results: $(\hat{\beta}_j - \beta_j)/\hat{\sigma}_{\hat{\beta}_j} \sim t_{n-p}$

- Consider a single-parameter estimator, $\hat{\beta}_j$, with standard deviation, $\sigma_{\hat{\beta}_j}$, given by the square root of element (j, j) of $\mathbf{V}_{\hat{\beta}} = \mathbf{R}^{-1}\mathbf{R}^{-\mathsf{T}}\sigma^2$.

- An unbiased estimator of $\mathbf{V}_{\hat{\beta}}$ is

$$\hat{\mathbf{V}}_{\hat{\beta}} = \mathbf{R}^{-1}\mathbf{R}^{-\mathsf{T}}\hat{\sigma}^2 = \frac{1}{\sigma^2}\mathbf{V}_{\hat{\beta}}\hat{\sigma}^2.$$

- An estimator of $\hat{\sigma}_{\hat{\beta}_j}$ is therefore given by the square root of element (j, j) of $\hat{\mathbf{V}}_{\hat{\beta}}$ and it is clear that

$$\hat{\sigma}_{\hat{\beta}_j} = \sigma_{\hat{\beta}_j}\hat{\sigma}/\sigma.$$

Linear model theory (Section 4 of the lecture notes)

Further inference results: $(\hat{\beta}_j - \beta_j)/\hat{\sigma}_{\hat{\beta}_j} \sim t_{n-p}$

$\hookrightarrow$ Hence, the pivotal statistic is

$$T = \frac{\hat{\beta}_j - \beta_j}{\hat{\sigma}_{\hat{\beta}_j}} = \frac{\hat{\beta}_j - \beta_j}{\sigma_{\hat{\beta}_j}\hat{\sigma}/\sigma} = \frac{(\hat{\beta}_j - \beta_j)/\sigma_{\hat{\beta}_j}}{\sqrt{\frac{1}{\sigma^2}\hat{\sigma}^2}} = \frac{(\hat{\beta}_j - \beta_j)/\sigma_{\hat{\beta}_j}}{\sqrt{\frac{1}{\sigma^2}\|\mathbf{r}\|^2/(n-p)}} \sim \frac{N(0,1)}{\sqrt{\chi^2_{n-p}/(n-p)}} \sim t_{n-p}$$

$\hookrightarrow$ This result can be used for any β_j exactly as it was used for the simple linear model (to test hypothesis and construct confidence intervals).

Linear model theory (Section 4 of the lecture notes)

Further inference results: testing $H_0 : \mathbf{C}\beta = \mathbf{d}$

- Suppose that $\mathbf{C}$ is a $q \times p$ matrix and $\mathbf{d}$ a $q \times 1$ dimensional vector, where $q < p$.
- Under H_0, we have $\mathbf{C}\hat{\beta} - \mathbf{d} \sim N(\mathbf{0}, \mathbf{C}\mathbf{V}_{\hat{\beta}}\mathbf{C}^{\mathsf{T}})$, from basic properties of the transformation of normal random vectors.
- Any positive definite matrix can be factored into the crossproduct of a triangular matrix with itself: a Cholesky decomposition.
- Forming the Cholesky decomposition $\mathbf{L}^{\mathsf{T}}\mathbf{L} = \mathbf{C}\mathbf{V}_{\hat{\beta}}\mathbf{C}^{\mathsf{T}}$ (if the design matrix $\mathbf{X}$ is full rank, then $\mathbf{V}_{\boldsymbol{\beta}}$ is positive definite), it is then easy to show that, under H_0,

$$\mathbf{L}^{-\mathsf{T}}(\mathbf{C}\hat{\beta} - \mathbf{d}) \sim N(\mathbf{0}, \mathbf{I})$$

- Therefore,

$$(\mathbf{C}\hat{\beta} - \mathbf{d})^{\mathsf{T}}(\mathbf{C}\mathbf{V}_{\hat{\beta}}\mathbf{C}^{\mathsf{T}})^{-1}(\mathbf{C}\hat{\beta} - \mathbf{d}) = (\mathbf{C}\hat{\beta} - \mathbf{d})^{\mathsf{T}}\mathbf{L}^{-1}\mathbf{L}^{-\mathsf{T}}(\mathbf{C}\hat{\beta} - \mathbf{d}) \sim \sum_{i=1}^{q} N(0,1)^2 \sim \chi^2_q.$$

Linear model theory (Section 4 of the lecture notes)

Further inference results: testing $H_0 : \mathbf{C}\beta = \mathbf{d}$

$\hookrightarrow$ As in the previous test, plugging in $\hat{\mathbf{V}}_{\hat{\beta}} = \mathbf{V}_{\hat{\beta}}\hat{\sigma}^2/\sigma^2$ gives the computable test statistic and its distribution under H_0:

$$
\begin{aligned}
F &= \frac{1}{q}(\mathbf{C}\hat{\beta} - \mathbf{d})^{\mathsf{T}}(\mathbf{C}\hat{V}_{\hat{\beta}}\mathbf{C}^{\mathsf{T}})^{-1}(\mathbf{C}\hat{\beta} - \mathbf{d}) \\
&= \frac{1}{q}(\mathbf{C}\hat{\beta} - \mathbf{d})^{\mathsf{T}}(\mathbf{C}\mathbf{V}_{\hat{\beta}}(\hat{\sigma}^2/\sigma^2)\mathbf{C}^{\mathsf{T}})^{-1}(\mathbf{C}\hat{\beta} - \mathbf{d}) \\
&= \frac{\sigma^2}{q\hat{\sigma}^2}(\mathbf{C}\hat{\beta} - \mathbf{d})^{\mathsf{T}}(\mathbf{C}V_{\hat{\beta}}\mathbf{C}^{\mathsf{T}})^{-1}(\mathbf{C}\hat{\beta} - \mathbf{d}) \\
&= \frac{(\mathbf{C}\hat{\beta} - \mathbf{d})^{\mathsf{T}}(\mathbf{C}V_{\hat{\beta}}\mathbf{C}^{\mathsf{T}})^{-1}(\mathbf{C}\hat{\beta} - \mathbf{d})/q}{\frac{1}{\sigma^2}\|\mathbf{r}\|^2/(n-p)} \\
&\sim \frac{\chi^2_q/q}{\chi^2_{n-p}/(n-p)} \sim F_{q,n-p}.
\end{aligned}
$$

$\hookrightarrow$ This result can be used to compute a p-value for the test.

Linear model theory (Section 4 of the lecture notes)

Further inference results: testing $H_0 : \mathbf{C}\beta = \mathbf{d}$

- When interest is in testing whether several elements of β could simultaneously be zero, the following result can be used:

$$F = \frac{(\text{RSS}_0 - \text{RSS}_1)/q}{\text{RSS}_1/(n-p)}.$$

- RSS_1 denotes the residual sum of squares for the full model, which contains p parameters.

- RSS_0 denotes the residual sum of squares for the reduced model, in which the H_0 is true. Let the reduced model contain p_0 parameters.

- Note that $q = p - p_0$.